

Robust Coordinated Distribution System Planning Considering Transactive DSO's Market

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Abstract—In this paper, a robust distribution system expansion planning (DSP) approach is presented to supply the load growth locally and move toward nearly zero energy local distribution areas (LDAs). In the proposed approach, a distribution system operator (DSO) is responsible for secure and optimum operation of LDAs. Therefore, investors on distribution system upgrades use this approach to maximize the profit on investments by determining the installation year of new distribution feeders and energy resources, distributed energy resource (DER) placements and sizes considered by corresponding DSOs. The accurate AC power flow solution is used and mathematical methods are developed to model the DSP as a quadratically constrained programming (QCP) problem. The Benders decomposition is applied to investigate the reliability and the optimality of the proposed plan and correspondingly modify the investment plan as required. The uncertainty of renewable DER forecast errors, locational marginal price (LMP) of an LDA at a transmission bus, and transactive power associated with a load bus is modeled using robust optimization. The proposed transactive DSP (TDSP) approach is implemented on the IEEE 33-bus distribution test system and the results are analyzed and validated. The proposed numerical results show the optimality and the robustness of the proposed approach.

Index Terms—Distribution system expansion planning, distribution system operator, near to zero local distribution area.

NOMENCLATURE

Indices and Sets

b	Index for load block.
C/E	Index for candidate/existing units or feeders.
f	Index for the feeders.
m, n, k	Index for distribution network nodes.
h	Index for DG units.
h_r	Index for renewable DG units.

s	Index for scenario.
t	Index for year.
$\sim$	Index for given variables.
$-$	Index for forecasted values of the uncertain parameters.
$\wedge$	Index for forecast error of the uncertain parameters.
D_N	Set of distribution network nodes.
D_A	Set of distribution lines.
<i>Parameters</i>	
d	Discount rate.
$PDG_{\max}$	Maximum generation capacity of non-renewable DERs.
$PDG_{\min}$	Minimum generation capacity of non-renewable DERs.
DT	Duration of load block.
H	Number of non-renewable DERs' units.
H_r	Number of renewable DERs' units.
$I^{\max}$	Limit on the feeders' current
IC	Annual investment cost of candidate DERs.
LMP	Forecasted LMP at LDA bus.
$LOEE$	Calculated $LOEE$ (loss of the energy expectation).
$LOEE_o$	Target $LOEE$.
N	Number of distribution network nodes.
N_b	Number of load blocks.
N_{fC}	Number of candidate feeders.
N_s	Number of scenarios.
OC	Operation cost of DERs' units.
PL/QL	Active/reactive load at the distribution node.
$PN_{\max}$	Upper bound limit on PN
$PN_{\min}$	Lower bound limit on PN
PR	Probability of a scenario.
RB_1	Robust budget associated with error on the output power of renewable DERs.
RB_2	Robust budget associated with forecast error on the market price (LMP).
RB_3	Robust budget associated with forecast error on the load power.
$SL_{\max}$	Maximum shiftable load.
$SL_{\min}$	Minimum shiftable load.
T	Number of planning years.
UX	A binary parameter which shows DG unit availability status.
UX_r	A binary parameter which shows renewable DG unit availability status.

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UY	A binary parameter which shows line availability status.
$v, \bar{v}$	Lower and upper bounds on voltage levels.
$x/r/Z$	Reactance/resistance/impedance of the feeders.
π	Price of feeders power flow.
<i>Variables</i>	
PDG	Generation power of non-renewable DERs.
RDG	Generation power of renewable DERs.
PF/QF	Active/reactive power flow of the feeders.
PN	Exchange power from the transmission network.
SL	Shifted load.
v	Voltage of distribution nodes.
X	Binary variables which indicate the installation status of DERs' units.
Xr	Binary variables which indicate the installation status of renewable DERs' units.
Y	Binary variables which indicate installation status of distribution line.
RDG	Generation power of renewable DERs.
$\bar{\lambda}, \underline{\lambda}, \bar{\mu}, \underline{\mu}, \bar{q}, \underline{q}, \bar{i}, \underline{i}, \bar{\omega}, \underline{\omega}$	Dual variables.
ω_1	A less than 1 positive variable to model the violation of renewable DERS production.
ω_2	A less than 1 positive variable to model the LMP violation.
ω_3	A less than 1 positive variable to model the Load power violation.
<i>Matrix and Vectors</i>	
RP_1	The matrix of positive unbalances power (RP_1).
RP_2	The matrix of negative unbalances power (RP_2).
RQ_1	The matrix of positive unbalance reactive power (RQ_1).
RQ_2	The matrix of negative unbalance reactive power (RQ_2).

I. INTRODUCTION

THE growing penetration of distributed energy resources (DERs) provides flexibility in the customer side of the electrical power system [1]–[3]. This rapid move toward active and near to zero local distribution areas (LDAs) provides several advantages for the power system such as improving the power system resiliency and efficiency [4], [5]. However, the improper operation and planning of these flexible DERs may create reliability and security issues for the LDAs as the most important parts of the power system to deliver electricity to the customers. Also, integrating of renewable DERs as the clean and low operating cost power generation resources increases the uncertainty of the power system which has a significant effect on the efficiency and reliability of the power system [6]–[8]. Therefore new enhanced methods are needed for secure and optimal

expansion planning and operation of the new active distribution network to supply annual growing active and reactive loads with minimum imposed cost while the reliable and secure operation of the LDA is satisfied [9]–[11].

To provide the proper platform and modify the planning and operation approaches regarding new smart and active distribution networks, a new independent distribution system operator (DSO) is considered in the literature which is responsible for optimal and secure operation and planning of its LDA [12], [13]. Each LDAs is connected to transmission network (TN) to exchange power with TN in the LMP of the joint transmission bus. Moreover, there are local DERs such as renewable DGs, energy-saving units (ESUs), and demand response customers in the LDA. These DERs in the LDA cause demand-side flexibility, as well as flexible generation and equipped distribution network with additional optimality option for locally supplying LDA's load in a lower cost [1]–[3]. For effective use of these flexible resources, a proper platform is required to provide a technical-economical resource management approach. Transactive Energy (TE) provides a market-based platform for this purpose. In a transactive retail market, the market participants (DERs) aim to maximize their profit due to their optimum scheduling for generation or demand management. As the DSO is responsible for the secure and minimum cost planning and operation of the LDA, it specifies the minimum cost of resources planning and scheduling based on receiving bids from DER owners and investors [14].

DSO supervises planning of the LDAs to ensure the long-term optimal and secure supply of the LDAs' growing load power. Considering new DERs in the DSP, defer or totally cancel the costly expansion generation and transmission planning of the bulk power system, which also provides economic and the environmental advantages for the customers by replacing high-cost inefficient peak generating units with efficient DGs, clean renewable DERs and reducing the transmission loss. On the other hand, DSP is a helpful solution for some important economical, technical, and political issues associated with the expansion planning of bulk power system [2], [15], [16].

The usual proposed DSP approaches in the literature aim to determine new energy resources planning with the minimum cost including investment and operation cost of the project [17], [18]. But, in the restructured environment, objective of the private investor is to maximize the planning profit. Therefore, a proper price signal is essential to investigate the best investment in terms of profitability. In our previous work [19], bulk power system planning coordinated with DSP has been studied, in which detailed transactive operation of LDA is not considered.

Therefore, this paper proposes a transactive DSP (TDSP) approach that focuses on accurate modeling of transactive operation of LDA. Also, a two-stage min-max formulation is presented to provide a precise robust formulation for uncertainty modeling. The platform in Fig. 1 is considered for interactions between DSOs and ISOs, and the LDAs' prosumers including DERs, micro grids (MGs), load aggregators (LAs), demand response aggregators (DRAs), GenCos, and large customers as DSO's transactive market participants in individual LDAs.

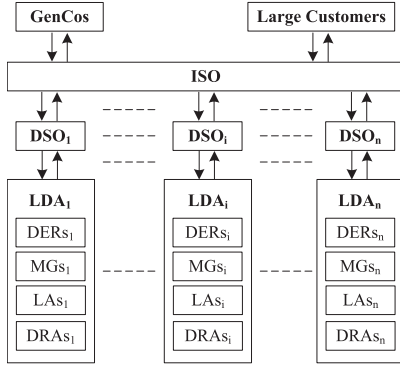


Fig. 1. Framework for the coordinated DSO market [2].

The proposed TDSP specifies the more profitable LDA's expansion planning while technical and security constraints of the network are satisfied. The modified Benders decomposition (MBD) is applied to provide an executable time-effective approach for the actual big local distribution network (LDN). The presented MBD includes the investment problem (IP) containing binary decision variables as the master problem and two sub-problems. The first sub-problem investigates the reliability of the investment solution. This paper uses "LOEE" as the reliability index that refers to the expected energy that in an electricity network and during a specific time cannot be supplied due to insufficient resources. The DSO's reliability sub-problem calculates LOEE of the DN, fixing the binary variables based on the master problem's solution and considering DER and feeder outages in different scenarios. If the master problem's solution satisfies the desired reliability level, the second sub-problem solve an optimality problem based on accurate AC load flow equations to calculate the price signal for the next iteration. Otherwise, the reliability sub-problem generates a new additional reliability cut for the next iteration of the IP. The presented TDSP is an iterative algorithm to calculate the proper distributed locational marginal prices (DLMPs) of the LDN nodes as the marginal cost of supplying load at different locations in each LDA. Therefore, DLMP is equivalent to the cost of supplying one kWh more demand at a specific place. The calculated DLMPs would be considered as the price signals for the most effective iterative solution. The stopping criteria of this iterative approach is satisfied when the difference between the calculated DLMP in two recent iterations is in the desired range of accuracy.

In summary, the contributions of the paper could be listed as follows:

- This paper proposes a transactive DSP (TDSP) approach that focuses on transactive operation of LDA to supply the load growth locally and move toward nearly zero energy local distribution areas (LDAs).
- The accurate AC power flow solution is used and mathematical methods are developed to model the DSP as a quadratically constrained programming (QCP) problem to consider the effects of the reactive power as well as the active power.

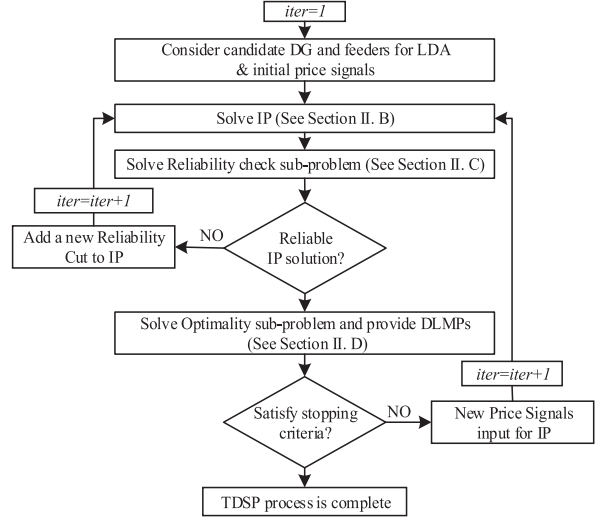


Fig. 2. Proposed Transactive DSP algorithm.

- The modified Benders decomposition is applied to investigate the reliability and the optimality of the proposed plan and correspondingly modify the investment plan as required.
- The proposed algorithm calculates LDA's DLMPs to improve the input price signals for the next iteration of the TDSP process.
- Also, a two-stage min-max formulation is presented to provide a precise robust formulation for uncertainty modeling.

The organization of the paper is as follows: Section II provides the formulation and the algorithm of the proposed transactive DSP approach. This section includes the formulations associated with IP as the master problem and the DSO's reliability check and optimality problem as the corresponding sub-problems. Numerical results are presented in Section III, using the IEEE 33 bus radial distribution system as the test system to analyze and show the effectiveness of the proposed TDSP approach. A brief conclusion is presented in Section IV.

II. FORMULATION OF TDSP

A. TDSP Algorithm

This paper considers the expansion planning for the smart restructured distribution networks in which the interactions are profit-based. Therefore, investors will interest on the projects with more economical gains. On the other hand, the reliable, secure, and cost-effective operation of the LDA are the most important DSO's objectives.

The proposed iterative TDSP algorithm shown in Fig. 2, investigates the most profitable solution as the planning decision to specify optimal places, sizes, and installation years of new DERs and feeders for the entire planning horizon while considering the reliable, secure and optimal operation of the LDA. The proposed approach utilizes the modified BD to solve the complicated planning problem dividing into a simple mixed-integer investment problem as the master problem and corresponding reliability and optimality problems. As the accurate AC power flow equations

are used to model DN's constraints, the sub-problems are formulated as a problem with quadratic constraints. Therefore, the proposed TDSP approach is capable to consider the effects of reactive power as well as active and reactive power losses on the final DSP decision.

In the first iteration of the TDSP algorithm, the master problem is solved considering desired initial values for the price signals. The proposed algorithm has two loops. The inner loop is associated with DSO's reliability check and generates corresponding cuts. The generated cut in each iteration is a new additional constraint for IP to conduct the solution toward an acceptable investment decision from the DN reliability point of view. As a reliable solution is provided, the TDSP algorithm would be continued to the next loop which improves the input price signals for the next iteration of the TDSP process. The outer loop solves the optimality problem considering the provided reliable solution from the output of the previous loop. Indeed, the optimality problem simulates the DSO's clearing process for distribution transactive market to calculate the modified DLMPs as the new price signals for the iterative TDSP approach. As the accurate AC power flow equations are utilized in this iterative procedure for improving the DLMPs, all affecting parameters such as reactive power consumption and the energy losses due to both active and reactive power losses are reflected in the price signals.

B. Investment Problem

In this section, the formulation and details of the IP are presented. The objective of the IP is to specify the optimum size and installation places of new DERs and upgrading the feeders to meet the growing LDA's load locally. Demand response is the other option to reduce the planning cost and helps to secure operation of the LDA in the situation of inadequate accessible energy resources or network congestion. This paper models the IP as a maximization problem which aims to maximize the total net present value (NPV) of the yearly planning profit all over the entire planning horizon. The annualized investment cost (IC) of the new DERs or feeders are considered to formulate the IP as follows:

$$\begin{aligned} \max \text{profit} &= \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(DLMP_{n,b,t} \cdot \left(\sum_{h=1}^H PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} RDG_{n,b,t}^{C,hr} \right) \right)}{(1+d)^{(t-1)}} \\ &+ \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{f_c=1}^{N_{f_c}} \frac{DT_{b,t} \cdot \left(\pi_{f_c,b,t} \cdot PF_{f_c,b,t}^C \right)}{(1+d)^{(t-1)}} \\ &- \sum_{t=1}^T \sum_{n=1}^N \sum_{h=1}^H \sum_{hr=1}^{H_r} \frac{(IC_{n,t}^h \cdot X_{n,t}^h + IC_{n,t}^{hr} \cdot X_{n,t}^{hr})}{(1+d)^{(t-1)}} \\ &- \sum_{t=1}^T \sum_{f_c=1}^{N_{f_c}} \frac{(IC_{f_c,t} \cdot Y_{f_c,t})}{(1+d)^{(t-1)}} \end{aligned}$$

$$\begin{aligned} &- \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(\sum_{h=1}^H OC_{n,b,t}^{C,h} \cdot PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} OC_{n,b,t}^{C,hr} \cdot RDG_{n,b,t}^{C,hr} \right)}{(1+d)^{(t-1)}} \\ &- \sum_{t=1}^T \sum_{b=1}^{N_b} \frac{DT_{b,t} \cdot LMP_{b,t} \cdot PN_{b,t}}{(1+d)^{(t-1)}} \end{aligned} \quad (1)$$

Subject to:

$$\begin{cases} X_{n,t}^h \leq X_{n,t+1}^h \\ X_{n,t}^{hr} \leq X_{n,t+1}^{hr}; \quad m, n \in \mathbf{D}_N^C \\ Y_{mn,t} \leq Y_{mn,t+1} \end{cases} \quad (2)$$

$$|PF_{mn,b,t}^C| \leq PF_{mn}^{\max} \cdot Y_{mn,t}^C; \quad m, n \in \mathbf{D}_N^C \quad (3)$$

$$\begin{cases} PDG_{n,\min}^{C,h} \cdot X_{n,t}^h \leq PDG_{n,b,t}^{C,h} \leq PDG_{n,\max}^{C,h} \cdot X_{n,t}^h \\ 0 \leq RDG_{n,b,t}^{C,hr} \leq R\overline{DG}_{n,b,t}^{C,hr} \cdot X_{n,t}^{hr} \end{cases}; \quad n \in \mathbf{D}_N^C \quad (4)$$

$$PN_{\min} \leq PN_{b,t} \leq PN_{\max} \quad (5)$$

Equation (1) is the objective function (OF) of the IP which represents the total NPV of the yearly gain due to investment on the new DERs and feeders upgrade over the entire planning horizon. The first terms in this OF are corresponding to the revenue of power generation of the new DERs including renewable DERs. For considering the profit due to feeder upgrades, we have supposed the revenue due to power flow in the feeders. This power flow revenue has been considered as a function of the flowed power (PF) from the feeder and the power flow price (π). Therefore, the second term in (1) reflects the revenue due to the power flow from the new feeders. The first part of the third term includes the annual investment cost on new DER, while the second part of the third term includes the annual investment cost on feeder upgrades. The operation costs of the planned DERs including renewable DERs are provided in the fourth term of the OF to ensure that new DERs are chosen from those candidate DERs which cause less operating cost. The last term in OF represents the cost of power exchange with transmission network which is settled in the LMP of the interfaced transmission bus to LDA. In our formulation, we use the net present value (NPV) to analyze the profitability of the investment by calculating the difference between the present value of cash inflows and cash outflows over the planning horizon. To calculate net present values, we have discounted the annual investment costs (ICs), operating costs, and the revenues due to power generation of the new DERs and the power flow from the new feeders at the discount rate d . Constraint (2) prevents to repeat an investment at year t in the following years. Eq. (3) shows the constraint on the maximum power flow of candidate feeders. Constraint (4) is associated with the minimum and maximum power generations of the candidate DGs. Equation (5) limits the minimum and maximum power exchanges of the LDA with the transmission network.

C. DSO's Reliability Check

To provide a reliable and technically acceptable investment decision on the new DERs and feeders, the proposed TDSP

includes a reliability check sub-problem. The objective of this sub-problem is to investigate the DN reliability of the IP solution to ensure reliable and secure operation of the LDA over the entire planning horizon. For this purpose, the reliability sub-problem is solved for the scenarios associated with the outage of DERs, feeders, or a combination of the two to calculate the LOEE as the LDA reliability index [17], [20]. The objective of the reliability problem is to minimize the power mismatch (both active and reactive parts) in each scenario. Therefore, positive variables $RP_{1,s}$, and $RP_{2,s}$ are introduced in (6) for the active power mismatch in each LDA's node which correspond with $RQ_{1,s}$, and $RQ_{2,s}$ for the reactive part of the power mismatch. The DSO's reliability check sub-problem are formulated as follows:

$$\begin{aligned} \text{Min } U_{b,t,s}^r = & \mathbf{1}^T * \mathbf{RP}_{1,s} + \mathbf{1}^T * \mathbf{RP}_{2,s} \\ & + \mathbf{1}^T * \mathbf{RQ}_{1,s} + \mathbf{1}^T * \mathbf{RQ}_{2,s} \end{aligned} \quad (6)$$

Subject to:

$$\begin{cases} UX_{n,t,s}^{C,h} \leq \tilde{X}_{n,t}^h \\ UX_{n,t,s}^{C,hr} \leq \tilde{X}_{n,t}^{hr} \\ UY_{mn,t,s}^C \leq \tilde{Y}_{mn,t} \end{cases} \quad m, n \in \mathbf{D}_N^C \quad (7)$$

$$\begin{aligned} & \sum_m (PF_{mn,b,t,s}^E + PF_{mn,b,t,s}^C) + \sum_{h=1}^H (PDG_{n,b,t,s}^{C,h} \\ & + RDG_{n,b,t,s}^{C,h}) + PN_{b,t,s} \\ & = PL_{n,b,t,s} + PSL_{n,b,t,s} + r_{mn,s}^E \cdot TI_{mn,b,t,s}^E + r_{mn,s}^C \cdot TI_{mn,b,t,s}^C \\ & + \sum_k (PF_{nk,b,t,s}^E + PF_{nk,b,t,s}^C) + RP_{1n,b,t,s} \\ & - RP_{2n,b,t,s}; m, n \in \mathbf{D}_N \end{aligned} \quad (8)$$

$$\begin{aligned} & \sum_m (QF_{mn,b,t,s}^E + QF_{mn,b,t,s}^C) + \sum_{h=1}^H (tg\varphi_{DG_n^{C,h}})^{-1} \\ & \cdot PDG_{n,b,t,s}^{C,h} + QN_{b,t,s} \\ & = (tg\varphi_{L_n})^{-1} \cdot (PL_{n,b,t,s} + PSL_{n,b,t,s}) + x_{mn,s}^E \cdot TI_{mn,b,t,s}^E \\ & + x_{mn,s}^C \cdot TI_{mn,b,t,s}^C \\ & + \sum_k (QF_{nk,b,t,s}^E + QF_{nk,b,t,s}^C) \\ & + RQ_{1n,b,t,s} - RQ_{2n,b,t,s}; m, n \in \mathbf{D}_N \end{aligned} \quad (9)$$

$$\begin{aligned} TV_{m,b,t,s} = & TV_{n,b,t,s} + 2 \cdot (r_{mn}^E \cdot PF_{mn,b,t,s}^E \\ & + x_{mn}^E \cdot QF_{mn,b,t,s}^E) \\ & - |Z_{mn}^E|^2 \cdot TI_{mn,b,t,s}^E; m, n \in \mathbf{D}_N^E \end{aligned} \quad (10)$$

$$\begin{aligned} & |TV_{m,b,t,s} - TV_{n,b,t,s} - 2 \cdot (r_{mn}^C \cdot PF_{mn,b,t,s}^C \\ & + x_{mn}^C \cdot QF_{mn,b,t,s}^C) + |Z_{mn}^C|^2 \cdot TI_{mn,b,t,s}^C| \\ & \leq M \cdot (1 - \tilde{Y}_{mn,t}); m, n \in \mathbf{D}_N^C (\underline{\lambda}_{mn,b,t,s}^C, \bar{\lambda}_{mn,b,t,s}^C) \end{aligned} \quad (11)$$

$$(v)^2 \leq TV_{m,b,t,s} \leq (\bar{v})^2; m \in \mathbf{D}_N \quad (12)$$

$$\begin{aligned} & (PF_{mn,b,t,s}^E)^2 + (QF_{mn,b,t,s}^E)^2 + \frac{(TV_{m,b,t,s} - TI_{mn,b,t,s}^E)^2}{4} \\ & \leq \frac{(TV_{m,b,t,s} + TI_{mn,b,t,s}^E)^2}{4}; m, n \in \mathbf{D}_N^E \end{aligned} \quad (13)$$

$$\begin{aligned} & (PF_{mn,b,t,s}^C)^2 + (QF_{mn,b,t,s}^C)^2 + \frac{(TV_{m,b,t,s} - TI_{mn,b,t,s}^C)^2}{4} \\ & \leq \frac{(TV_{m,b,t,s} + TI_{mn,b,t,s}^C)^2}{4}; m, n \in \mathbf{D}_N^C \end{aligned} \quad (14)$$

$$|PF_{mn,b,t,s}^E| \leq PF_{mn}^{\max} \cdot UY_{mn,t,s}^E; m, n \in \mathbf{D}_N^E \quad (15)$$

$$|QF_{mn,b,t,s}^E| \leq QF_{mn}^{\max} \cdot UY_{mn,t,s}^E; m, n \in \mathbf{D}_N^E \quad (16)$$

$$TI_{mn,b,t,s}^E \leq (I_{mn}^{\max})^2 \cdot UY_{mn,t,s}^E; m, n \in \mathbf{D}_N^E \quad (17)$$

$$\begin{aligned} & |PF_{mn,b,t,s}^C| \leq PF_{mn}^{\max} \cdot UY_{mn,t,s}^C \\ & \cdot \tilde{Y}_{mn,t}; m, n \in \mathbf{D}_N^C (\underline{\mu}_{mn,b,t,s}^C, \bar{\mu}_{mn,b,t,s}^C) \end{aligned} \quad (18)$$

$$\begin{aligned} & |QF_{mn,b,t,s}^C| \leq QF_{mn}^{\max} \cdot UY_{mn,t,s}^C \\ & \cdot \tilde{Y}_{mn,t}; m, n \in \mathbf{D}_N^C (\underline{q\mu}_{mn,b,t,s}^C, \bar{q\mu}_{mn,b,t,s}^C) \end{aligned} \quad (19)$$

$$\begin{aligned} & TI_{mn,b,t,s}^C \leq (I_{mn}^{\max})^2 \cdot UY_{mn,t,s}^C \\ & \cdot \tilde{Y}_{mn,t}; m, n \in \mathbf{D}_N^C (\underline{i\mu}_{mn,b,t,s}^C, \bar{i\mu}_{mn,b,t,s}^C) \end{aligned} \quad (20)$$

$$\begin{aligned} & PDG_{n,\min}^{C,h} \cdot UX_{n,t,s}^{C,h} \cdot \tilde{X}_{n,t}^h \\ & \leq PDG_{n,b,t,s}^{C,h} \leq PDG_{n,\max}^{C,h} \cdot UX_{n,t,s}^{C,h} \cdot \tilde{X}_{n,t}^h; \\ & n \in \mathbf{D}_N (\underline{\omega}_{n,b,t,s}^{C,h}, \bar{\omega}_{n,b,t,s}^{C,h}) \end{aligned} \quad (21)$$

$$\begin{aligned} & 0 \leq RDG_{n,b,t,s}^{C,hr} \leq \overline{RDG}_{n,b,t}^{C,hr} \cdot UX_{n,t,s}^{C,hr} \cdot \tilde{X}_{n,t}^{hr}; \\ & n \in \mathbf{D}_N (\underline{\omega}_{n,b,t,s}^{C,hr}, \bar{\omega}_{n,b,t,s}^{C,hr}) \end{aligned} \quad (22)$$

$$PN_{\min} \leq PN_{b,t} \leq PN_{\max} \quad (23)$$

$$SL_{n,\min} \leq SL_{n,b,t,s} \leq SL_{n,\max}; n \in \mathbf{D}_N \quad (24)$$

$$\sum_{b=1}^{N_b} SL_{n,b,t,s} = 0; n \in \mathbf{D}_N \quad (25)$$

The following auxiliary variables TI_{mn} , TV_n are introduced to present a convex formulation in the form of a second-order cone programming (SOCP) [21]:

$$\begin{cases} TI_{mn}^E = (I_{mn}^E)^2; m, n \in \mathbf{D}_N^E \\ TI_{mn}^C = (I_{mn}^C)^2; m, n \in \mathbf{D}_N^C \end{cases} \quad (26)$$

$$TV_m = |v_m|^2; m \in \mathbf{D}_N^i \quad (27)$$

The provided formulation of the DSO's reliability check sub-problem is in the form of the quadratically constrained programming (QCP). So, a commercial solver such as CPLEX could be employed in GAMS to provide the guaranteed optimal solutions. Constraint (7) indicates that only the planned DERs and feeders are considered for generating outage scenarios. Eqs.

(8), and (9) represent the active and reactive power balance constraints in each LDA's node, respectively. The equations that give the relationship between the voltage of starting and ending nodes of the existing feeders with their currents and the voltage drop on the feeders are presented in (10), which correspond with (11) for the planned feeders. Eq. (12) limits the nodes voltage. Eqs. (13) and (14) represent the SOCP form of the relationship between the active and reactive power flows, voltage of the starting node, and currents of the existing and planned feeders, respectively. Eqs. (15)–(17) show the constraints on the active and reactive power flows, and the current limits of the existing feeders, which correspond with Eqs. (18)–(20) for planned feeders. Eq. (21) represents up and down limits of the generating power of the planned DGs. Constraint (23) is associated with the limits on the LDA's exchange power with the transmission network. Equations (24) and (25) represent the constraints on the shifted load in the demand response program. Moreover, equation (25) indicates that the reduced load of each load block b in year t should be supplied in the other period times.

Once the DSO's reliability check sub-problem is solved, the equation (28) would be applied to calculate the total LOEE of the LDA. If the calculated $LOEE$ exceeds the considered LDA reliability requirement ($LOEE_0$), the reliability Benders cut would be generated. Considering the idea of [7], the valid Benders cut to ensure reliability requirements in the planning problem, has been derived as (29) and added to IP as a new constraint for the next Bender iteration.

$$LOEE = \sum_{n=1}^N \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{s=1}^{N_s} PR_s \cdot \frac{RP_{1n,b,t,s} + RP_{2n,b,t,s}}{PL_{n,b,t,s}} \quad (28)$$

$$\begin{aligned} & \sum_{s=1}^{N_s} PR_s \cdot (RP_{1n,b,t,s} + RP_{2n,b,t,s}) + \sum_{s=1}^{N_s} \sum_{dl_C=1}^{N_{dl_C}} \\ & \times PR_s \left\{ \begin{aligned} & -M \cdot (Y_{mn,t} - \tilde{Y}_{mn,t}) \\ & \cdot (\underline{\lambda}_{mn,b,t,s}^C + \bar{\lambda}_{mn,b,t,s}^C) \\ & + \widetilde{UY}_{mn,t,s}^C \cdot (Y_{mn,t} - \tilde{Y}_{mn,t}) \\ & \cdot \left[\begin{aligned} & PF_{mn}^{C,\max} \cdot (\underline{\mu}_{mn,b,t,s}^C + \bar{\mu}_{mn,b,t,s}^C) \\ & + QF_{mn}^{C,\max} \cdot (\underline{q}_{mn,b,t,s}^C + \bar{q}_{mn,b,t,s}^C) \\ & + (I_{mn}^{\max})^2 \cdot (\underline{i}_{mn,b,t,s}^C + \bar{i}_{mn,b,t,s}^C) \end{aligned} \right] \end{aligned} \right\} \\ & + \sum_{s=1}^{N_s} \sum_{n=1}^N PR_s \left\{ \begin{aligned} & (Xr_{n,t}^{hr} - \tilde{X}r_{n,t}^{hr}) \cdot \widetilde{UX}r_{n,t,s}^{hr} \\ & \cdot \widetilde{RDG}_{n,b,t}^{C,hr} \cdot \widetilde{\omega}r_{n,b,t,s}^{C,hr} \\ & + (X_{n,t}^h - \tilde{X}_{n,t}^h) \cdot \widetilde{UX}_{n,t,s}^h \\ & \cdot \left[\begin{aligned} & PDG_{n,b,t}^{C,\max} \cdot \widetilde{\omega}_{n,b,t,s}^{C,h} \\ & - PDG_{n,b,t}^{C,\min} \cdot \underline{\omega}_{n,b,t,s}^{C,h} \end{aligned} \right] \end{aligned} \right\} \\ & \leq LOEE_0 \cdot \sum_{s=1}^{N_s} PR_s \cdot PL_{n,b,t,s} \quad (29) \end{aligned}$$

D. DSO's Optimality Problem

If IP solution provides an acceptable reliability for the LDA, the TDSP algorithm would be continued with the DSO's optimality sub-problem (OSP). This OSP solves an AC optimal power flow in the LDA to determine the new values of $DLMP_{n,b,t}$ and $\pi_{mn,b,t}$ for the next iteration of the proposed iterative TDSP approach. The normal operating condition is considered and the IP solution is applied for this purpose. The formulation of the OP is presented as follows:

$$\begin{aligned} & \min \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N DT_{b,t} \cdot \left(\frac{DLMP_{n,b,t} \cdot \left(\sum_{h=1}^H PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} RDG_{n,b,t}^{C,hr} \right)}{(1+d)^{(t-1)}} \right) \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{fc=1}^{N_{fc}} \frac{DT_{b,t} \cdot (\pi_{fc,b,t} \cdot PF_{fc,b,t}^C)}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(\sum_{h=1}^H OC_{n,b,t}^{C,h} \cdot PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} OC_{n,b,t}^{C,hr} \cdot RDG_{n,b,t}^{C,hr} \right)}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \frac{DT_{b,t} \cdot LM_{P_{b,t}} \cdot PN_{b,t}}{(1+d)^{(t-1)}} \quad (30) \end{aligned}$$

Subject to:

$$\begin{aligned} & \sum_m (PF_{mn,b,t}^E + PF_{mn,b,t}^C) + \sum_{h=1}^H \left(PDG_{n,b,t}^{C,h} \right. \\ & \left. + RDG_{n,b,t}^{C,h} \right) + PN_{b,t} \\ & = PL_{n,b,t} + PSL_{n,b,t} + r_{mn,s}^E \cdot TI_{mn,b,t}^E + r_{mn}^C \cdot TI_{mn,b,t}^C \\ & + \sum_k (PF_{nk,b,t}^E + PF_{nk,b,t}^C) - RL_{n,b,t}^i; m, n \in \mathbf{D_N}(\beta_{n,b,t}) \quad (31) \end{aligned}$$

$$\begin{aligned} & \sum_m (QF_{mn,b,t,s}^E + QF_{mn,b,t,s}^C) + \sum_{h=1}^H (tg\varphi_{DG_n^{C,h}})^{-1} \\ & \cdot PDG_{n,b,t,s}^{C,h} + QN_{b,t,s} \\ & = (tg\varphi_{L_n})^{-1} \cdot (PL_{n,b,t,s} + PSL_{n,b,t,s}) \\ & + x_{mn,s}^E \cdot TI_{mn,b,t,s}^E + x_{mn,s}^C \cdot TI_{mn,b,t,s}^C \\ & + \sum_k (QF_{nk,b,t,s}^E + QF_{nk,b,t,s}^C) - (tg\varphi_{L_n})^{-1} \\ & \cdot RL_{n,b,t,s}; m, n \in \mathbf{D_N}(\delta_{n,b,t}) \quad (32) \end{aligned}$$

Equations (10)–(25), for only normal operating condition ($s = 0$; $UY_{mn,t,s}^E = UX_{mn,t,s}^E = UY_{mn,t,s}^C = UX_{mn,t,s}^C = 1 \forall m, n, t$).

Subsequently, the new values of $DLMP_{n,b,t}$ and $\pi_{fc,b,t}$ are calculated as follows:

$$DLMP_{n,b,t}^i = (\beta_{n,b,t} + (tg\varphi_{L_n}^i) \cdot \delta_{n,b,t}) / DT_{b,t} \quad (33)$$

$$\pi_{mn,b,t} = DLMP_{m,b,t} - DLMP_{n,b,t} \quad (34)$$

where, $\beta_{n,b,t}$ and $\delta_{n,b,t}$ are dual variables of Eqs. (31) and (32), respectively, which reflect the portion of both active and reactive

power of the incremental load in the incremental supply cost. To economically encourage investors for investing in the places with more congestion, the power flow price for the new feeder between nodes m, n ($\pi_{mn,b,t}$) has been considered as (34), that is equal with difference between DLMPs of sending and receiving bus.

E. Uncertainty Modeling

In this section, the uncertainty modeling in our formulation is presented. The uncertainties associated with forecast errors on the renewable DGs power, the LMP of the transmission bus of the LDA, and the demanded load of the LDA have been considered. As there is no available accurate data on the probability density function of the uncertain parameters in the most power system problems, the robust optimization approach is applied for this purpose which models uncertainty through corresponding uncertainty sets, and does not need these kinds of data. The robust optimization approach considers the worst condition for the uncertain parameter to guarantee the optimal solution of the planning in all possible violation of the uncertain parameters. Considering the worst condition for uncertain parameters, impose a max-min optimization. The inner optimization, address a minimization on the uncertain parameters as decision variables, and the outer optimization, maximize the benefit of the decision for the resulted worst condition of the uncertain parameters. In our study, the uncertain parameters are introduced as follows:

$$RDG_{n,b,t}^{C,hr} = \overline{RDG}_{n,b,t}^{C,hr} \pm \omega_{1n,b,t}^{C,hr} \cdot \hat{RDG}_{n,b,t}^{C,hr} \quad (35)$$

$$LMP_{b,t} = \overline{LMP}_{b,t} \pm \omega_{2b,t} \cdot \hat{LMP}_{b,t} \quad (36)$$

$$PL_{n,b,t} = \overline{PL}_{n,b,t} \pm \omega_{3n,b,t} \cdot \hat{PL}_{n,b,t} \quad (37)$$

$$0 \leq \omega_{1n,b,t}^{C,hr} \leq 1; \sum_{n=1}^N \sum_{hr=1}^{H_r} \sum_{b=1}^{N_b} \sum_{t=1}^T \omega_{1n,b,t}^{C,hr} = RB_1 \quad (38)$$

$$0 \leq \omega_{2b,t} \leq 1; \sum_{b=1}^{N_b} \sum_{t=1}^T \omega_{2b,t} = RB_2 \quad (39)$$

$$0 \leq \omega_{3n,b,t} \leq 1; \sum_{n=1}^N \sum_{b=1}^{N_b} \sum_{t=1}^T \omega_{3n,b,t} = RB_3 \quad (40)$$

where, $\omega_{1n,b,t}^{C,hr}$, $\omega_{2b,t}$, $\omega_{3n,b,t}$, are associated with the forecast error on the renewable generation, $LMP_{b,t}$, and the load power, respectively with their corresponding robust budget as RB_1 , RB_2 , RB_3 . For risk-taking investment planning, these robust budgets are set to zero. Increasing the robust budget in the DSP, is associated with more robust decisions which reduces the risk of the planning due to the violations of the uncertain parameters from their forecasted values. Therefore, the maximum robust budget setting ($RB = 100\%$) is preferred for a risk-averse robust DSP. Robustness of the DSP imposes the robust counterparts (RCs) as the additional cost of the DSP problem and subsequently reduces the profit of the decision.

To specify the right operator ($\pm$) which is corresponding to the worst condition of the uncertain parameter violations from their forecasted values, the effects of the uncertain parameters changes on the cost or benefit of the decision are investigated as follows:

Through the linearity of the objective function (1) and the impressive constraints of the renewable DG power (the linearized form of active and reactive power balance presented in Eqs. (8), (9)), it is obvious that decreasing renewable generating power, decreases the profit and increases the cost (operating cost of renewable generation is very small compared to that of other resources). Therefore, the worst condition for the renewable generating power is associated with the subtraction operator in (35). As such, increasing the LMP increases the cost of the planning, the worst condition for $LMP_{b,t}$ is associated with summation operator in (36). Considering equations (8), (9), it is obvious that the effect of the load power forecast error on the planning profit is contrary to the effect of the forecast error on the renewable generating power. Therefore, the worst condition for the load power is associated with the summation operator in (37). The load increment from the forecasted value increases the generation cost. This incremental cost is calculated as the increased load power multiplied by the corresponding DLMP and the duration time (DT) of the load block. Considering the above elucidation, the formulation of the proposed problem is modified to consider the effects of the forecast error on the uncertain parameters and provide a robust TDSP. The provided TDSP algorithm in Fig. 2 is unchanged. However, to consider uncertainty modeling in this algorithm, the provided formulations in II-B to II-D are replaced with the modified formulations as follows:

Modified Investment Problem

To apply robust optimization, the investment problem is modified as follows:

$$\begin{aligned} & \max_{\omega} \min_{\omega} MOF \\ & = \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(D LMP_{n,b,t} \cdot \left(\sum_{h=1}^H PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} RDG_{n,b,t}^{C,hr} \right) \right)}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{f_C=1}^{N_{f_C}} \frac{DT_{b,t} \cdot (\pi_{f_C,b,t} \cdot PF_{f_C,b,t}^C)}{(1+d)^{(t-1)}} \\ & - \left\{ \begin{aligned} & + \sum_{t=1}^T \sum_{n=1}^N \sum_{h=1}^H \sum_{hr=1}^{H_r} \frac{(IC_{n,t}^h \cdot X_{n,t}^h + IC_{n,t}^{hr} \cdot X_{n,t}^{hr})}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{f_C=1}^{N_{f_C}} \frac{(IC_{f_C,t} \cdot Y_{f_C,t})}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(\sum_{h=1}^H OC_{n,b,t}^{C,h} \cdot PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} OC_{n,b,t}^{C,hr} \cdot RDG_{n,b,t}^{C,hr} \right)}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \frac{DT_{b,t} \cdot \overline{LMP}_{b,t} \cdot \overline{PL}_{b,t}}{(1+d)^{(t-1)}} \\ & + RC_1 + RC_2 + RC_3 \end{aligned} \right\} \quad (41) \end{aligned}$$

Subject to:

$$RC_1 = \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \sum_{hr=1}^{H_r} \frac{DT_{b,t} \cdot ((DLMP_{n,b,t} - OC_{n,b,t}^{C,hr}) \cdot \omega_{1n,b,t}^{C,hr} \cdot RDG_{n,b,t}^{C,hr})}{(1+d)^{(t-1)}} \quad (42)$$

$$RC_2 = \sum_{t=1}^T \sum_{b=1}^{N_b} \frac{DT_{b,t} \cdot PN_{b,t}^* \cdot \omega_{2b,t} \cdot LMP_{b,t}}{(1+d)^{(t-1)}} \quad (43)$$

$$RC_3 = \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot (DLMP_{n,b,t} \cdot \omega_{3n,b,t} \cdot LP_{n,b,t})}{(1+d)^{(t-1)}} \quad (44)$$

$$0 \leq \omega_{1n,b,t}^{C,hr} \leq X_{r,n,t}^{hr} \quad (45)$$

Equations (2)–(5), (38)–(40)

To prevent the nonlinearity term of multiplying two variables $PN_{b,t}$ and $\omega_{2b,t}$ in the model, calculated value of the exchanging power with the transmission network ($PN_{b,t}^*$) extracted from the previous iteration of the proposed iterative TDSP algorithm in Section II-A is used to provide a linear equation in (43).

As the inner optimization is the minimization of the negative terms, it could be rewritten as the maximization of the positive terms as $\max\{RC_1 + RC_2 + RC_3\}$, and the presented max-min optimization in (41), would be rewritten as a max-max optimization. This max-max optimization is the same as a single maximization problem with its modified objective function (MOF). Therefore, the final modified investment problem is rewritten as follows:

$$\begin{aligned} \max MOF = & \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(DLMP_{n,b,t} \cdot \left(\sum_{h=1}^H PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} RDG_{n,b,t}^{C,hr} \right) \right)}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{f_C=1}^{N_{f_C}} \frac{DT_{b,t} \cdot (\pi_{f_C,b,t} \cdot PF_{f_C,b,t}^C)}{(1+d)^{(t-1)}} \\ & - \left\{ \begin{aligned} & + \sum_{t=1}^T \sum_{n=1}^N \sum_{h=1}^H \sum_{hr=1}^{H_r} \frac{(IC_{n,t}^h \cdot X_{n,t}^h + IC_{n,t}^{hr} \cdot X_{r,n,t}^{hr})}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{f_C=1}^{N_{f_C}} \frac{(IC_{f_C,t} \cdot Y_{f_C,t})}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \sum_{n=1}^N \frac{DT_{b,t} \cdot \left(\sum_{h=1}^H OC_{n,b,t}^{C,h} \cdot PDG_{n,b,t}^{C,h} + \sum_{hr=1}^{H_r} OC_{n,b,t}^{C,hr} \cdot RDG_{n,b,t}^{C,hr} \right)}{(1+d)^{(t-1)}} \\ & + \sum_{t=1}^T \sum_{b=1}^{N_b} \frac{DT_{b,t} \cdot \overline{LMP}_{b,t} \cdot PN_{b,t}^*}{(1+d)^{(t-1)}} - (RC_1 + RC_2 + RC_3) \end{aligned} \right\} \quad (46) \end{aligned}$$

Subject to: (42)–(45), (2)–(5), (38)–(40).

Modified DSO's Reliability Check

To apply robust optimization in Subsection II-C, the only needed change to the DSO's reliability check problem is to

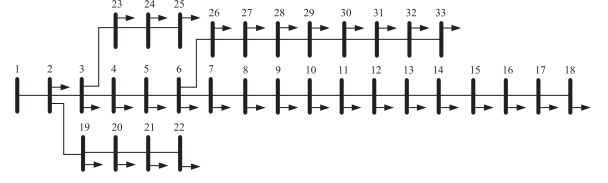


Fig. 3. LDA1 one-line diagram [23].

TABLE I
CANDIDATE DG DATA

Unit	Capacity (KVA)	FOR	O&MC (\$/MWh)	Annual IC (\$/MW)
H _{1,2}	300	0.015	6.5	55900
H _{r1,2}	150	0.008	1.6	98800

replace $PL_{n,b,t}$ in equations (8), (9) with following one:

$$PL_{n,b,t} = \overline{PL}_{n,b,t} + \omega_{3n,b,t}^* \cdot \hat{PL}_{n,b,t} \quad (47)$$

where, $\omega_{3n,b,t}^*$ is the calculated $\omega_{3n,b,t}$ in modified investment problem.

Modified DSO's Optimality Problem

To modify Subsection II-D, it is needed to replace uncertain parameters with the calculated ones based on considered robust budgets. Therefore, following equations are added to DSO's optimality problem to modify it for applying robust optimization:

$$RDG_{n,b,t}^{C,hr} = \overline{RDG}_{n,b,t}^{C,hr} - (\omega_{1n,b,t}^{C,hr})^* \cdot \hat{RDG}_{n,b,t}^{C,hr} \quad (48)$$

$$LMP_{b,t} = \overline{LMP}_{b,t} + (\omega_{2b,t})^* \cdot \hat{LMP}_{b,t} \quad (49)$$

$$PL_{n,b,t} = \overline{PL}_{n,b,t} + (\omega_{3n,b,t})^* \cdot \hat{PL}_{n,b,t} \quad (50)$$

where, $(\omega_{1n,b,t}^{C,hr})^*$, $(\omega_{2b,t})^*$, and $(\omega_{3n,b,t})^*$ are calculated $\omega_{1n,b,t}^{C,hr}$, $\omega_{2b,t}$, and $\omega_{3n,b,t}$ in modified investment problem, respectively.

This modified formulation of the TDSP problem is easily solvable by general programming tools such as GAMS in a single optimization problem. After solving the modified optimization problem, the robust profit of the planning is calculated as follows:

$$MOF^* - 2 \cdot RC_1^* - 2 \cdot RC_2^* - RC_3^* \quad (51)$$

where, stored parameters represent the calculated corresponding variables in modified investment problem.

III. NUMERICAL RESULTS

The 33-bus IEEE radial distribution network shown in Fig. 3 is used as the LDA to validate the proposed transactive DSP in a 10-years planning horizon. There are 32 load points and 32 distribution feeders in the simulated LDA which its data are available in [8], [22]. The data on the candidate DGs (including renewable DGs) are presented in Table I. The candidate feeders are assumed in parallel with the existing ones. The data for the candidate feeders are assumed to be the same as that of the existing ones. The forced outage rates (FORs) of existing and candidate feeders are 0.003, and 0.001, respectively.

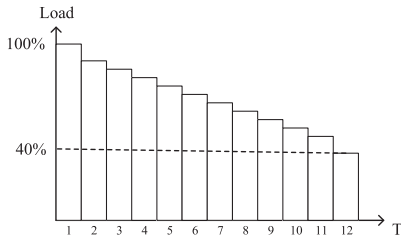


Fig. 4. Annual LDC.

In the starting year of the planning horizon (year 0), the active and reactive peak load of the LDA are 3715 kW, and 2300 kVAR, respectively. The assumption for the average yearly growth of both active and reactive load is 5%. The assumed annual discount rate in our numerical results is 5% ($d = 5\%$). To produce the load duration curve (LDC), 12 load block has been considered in each planning year in which the active and reactive load power changes between 40% and 100% of their annual peak values (see Fig. 4). In our planning study, a detailed load duration curve is not considered. However, load blocks has been considered to model transactive operation more accurately. Duration of each load block is large enough that we didn't need to consider detailed temporal constraints such as ramp constraints of DG units. In fact, load blocks divide each planning year t to several time periods to consider several peak load in one year instead considering only one peak load for each year. This approach enable us to consider DR, and other DERs scheduling in the planning problem. Considering resources available at the operating stage could provide more options for DSP to optimize planning decisions, as well as, ensuring feasible decisions. However, detailed temporal constraints are not considered. Moreover, in the published articles the detailed temporal constraints are not taken into account in long term planning problem. For instance, [24]–[28] investigate planning problem in monthly, seasonal or even annual basis where precise temporal constraints are not taken into account. Indeed, the hourly operation conditions in the planning horizon are grouped in larger duration of each load block. The considered monthly load blocks are the samples with the highest value of hourly demand for each month which are considered for all hours of the month. To maintain the chronological between load and renewable generation data, the renewable generation associated with the same monthly peak load sample is used in each block. Also, to provide a robust decision, the worst operating conditions based on considered robust budgets are investigated to guarantee a robust solution in all possible operating conditions. Therefore, detailed hourly or shorter period of changes of the renewable outputs and demand power and the possible temporal correlation between them are not the case in our long term robust study. While considering smaller durations for each load block may improve the accuracy of the model, the computational time will considerably be increased. However, in long term planning, decision makers try to make decisions on investing for years ahead in order to meet forecasted future electricity demand. In such a model, an accurate short term operation studies that model all detailed operation constraints are not considered to

make a tradeoff between accuracy and computation time. In our numerical study, 12 monthly peak load is considered in each year to improve the accuracy. For example, parameter $PL_{n,b,t}$ denotes active load power in n th node of distribution network at b th load block of planning year t . Therefore, $PL_{2,4,5}$ represents the active load power in node 2 at 4th month of the 5th planning year. 10% of the LDA's load is considered as DR. In the reliability sub-problem, the LOEE is calculated by considering scenarios with higher probability (scenarios with probability more than $1E-6$) as the reliability index of the LDA.

In the proposed modeling, the storage facilities are not considered as a separate DER in the formulations. However, our model is capable to consider the effect of the storage systems on the planning procedure by means of updating the constraint on the available generation of DERs with energy saving unit (ESU). For example, a generating units or a demand response customer with ESU has more capability and flexibility for gaining from local distribution market that can be reflected in the limits of corresponding constraints. For instance, a DER with ESU is able to save energy (charging) in low cost durations and sell saved energy (discharging) in high cost durations, as well as, its generation. Consequently, the capacity of DERs with storage is higher than those without storage to take advantage of the transactive market. This capability can be modeled by decreasing/increasing the minimum/maximum generation limits (PDG_{min}/PDG_{max}) for DERs with ESU based on their storage capacity. With the similar rationale, for demand response customer, the effect of ESU could be considered by decreasing/increasing the limits on the minimum/maximum shiftable load (SL_{min}/SL_{max}) in the corresponding constraints.

In our study, two stopping criteria has been considered. The inner iterative loop applies the benders' decomposition (BD) technique that provide a guaranteed convergence. The stopping criteria for the inner reliability loop, is that the calculated LOEE be equal or less than 1%. However, the stopping criteria of the outer DLMP loop is satisfied when the difference between the calculated DLMP in two recent iterations is in the acceptable range of accuracy. As the accurate AC power flow equations are applied in DLMP iterative loop, all affecting parameters such as reactive power consumption and the energy losses due to both active and reactive power losses are reflected in the price signals. In our numerical study, the best decision is the most profitable decision in which the operating cost, and the energy losses due to both the active and the reactive power consumption is optimized. Therefore, as the best decisions are reached the violation of the calculated DLMPs in the last two iterations would be in the desirable range (less than 5%).

To consider the effects of the uncertainties on the proposed transactive DSP approach, 3 case studies with different values (0%, 50%, and 100%) for the robust budgets are considered, and the corresponding results are presented in Table II. The zero robust budget is equivalent to deterministic approach in which all the parameters are completely known and the uncertainty is not considered. On the other hand, in a different case study, we have set the robust budget equal to 100%. The maximum robust budget is equivalent to risk-averse approach in which the worst condition has been considered in term of uncertain parameters.

TABLE II
PLANNING PROFIT AND RELIABILITY RESULTS FOR DIFFERENT RB

RB	Profit (Thousand \$)	LOEE (%)	Feeders Upgrade	Upgrade year
0%	1852.005	0.934	-	-
50%	1657.952	0.979	-	-
100%	1579.956	0.971	1-2	1

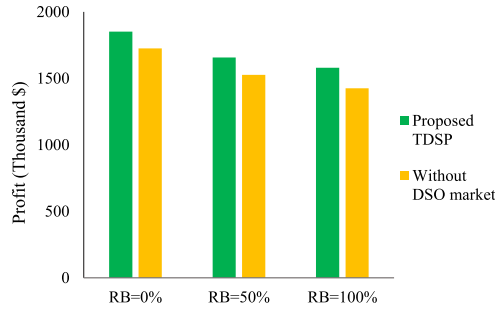


Fig. 5. Comparison chart between with and without transactive DSO market modeling.

TABLE III
GENERATING UNITS INSTALLATION YEAR

Nodes	n_7	n_{13}	n_{14}	n_{18}	n_{24}	n_{25}	n_{29}	n_{30}	n_{31}
H_1	8	-	3	-	-	6	7	2	-
H_2	-	-	5	-	-	9	-	4	-
H_{r1}	-	7	-	4	2	-	-	-	1
H_{r2}	-	-	-	5	8	-	-	-	1

As a conclusion from this table, increase in the robust budget of the TDSP problem which is corresponded with a more robust decision against uncertainties, increases the planning cost of the decision. Moreover, it is required to upgrade the feeder between nodes 1 and 2 in the first year in the case with maximum RBs to overcome the effects of the uncertainties on the reliability and security of the TDSP solution.

To clear the benefit of the proposed TDSP approach, in term of implementing a competitive profit-based planning and improving the profitability in the planning procedure, we have considered new case studies in which the DSO market is not modeled. The comparative chart between proposed TDSP and the planning approaches without DSO market modeling is presented in Fig. 5. From this figure, we conclude that proposed TDSP approach improves the profitability of the planning decisions by enabling a competitive transactive DSO market.

The following results are associated with considering the maximum robust budget in the simulations to guarantee the robustness of the TDSP solution. Table III presents the planned candidate DERs (including the renewable DERs) and their installation years considering DR = 10%. In this case, the total capacity of planned DERs' and the net present value of the total DERs investment cost are 3.45 MVA, \$1.051 million.

Table IV shows the results for different levels of demand response to investigate the effect of DR on the TDSP solution. As a result, from this table, increase in the DR level corresponds with lower planning cost or more profitable decisions which are more interesting for the investors. In the case with no DR, the

TABLE IV
PLANNING PROFIT AND RELIABILITY RESULTS FOR DIFFERENT DR

DR	Profit (Thousand \$)	LOEE (%)	Feeders Upgrade	Upgrade year
0%	841.069	0.915	1-2, 2-3	1
10%	1579.956	0.971	1-2	1
20%	1686.421	0.954	-	-

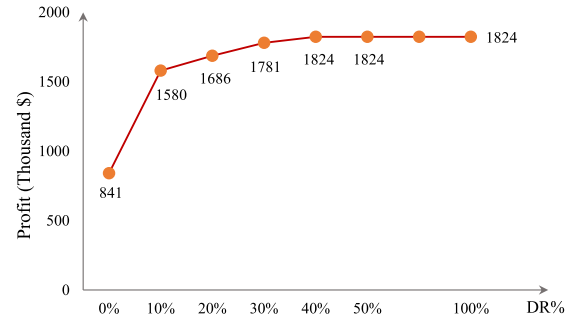


Fig. 6. The effect of DR on the planning profitability.

TDSP decision is the least profitable project which is associated with the least interesting planning project for the investors. In the case with 20% DR availability, the investment on the feeder upgrading would no longer be required. Therefore, considering the advantage of the transactive market in a smart DN is helpful to improve the investment pattern of TDSP toward a more efficient pattern with less investing on the new DN infrastructures which are high time and cost consuming projects.

To study the effect of DR percentage on the planning profitability as a critical parameter in our study, we have done a sensitivity analyses on this parameter by extending the second case study to higher DR percentage. The obtained results is provided in Fig. 6. This figure shows that by increasing DR percentages, the planning profit increases to about 1823 thousand dollar. Applying the first 10% DR has the most effect on the planning profit. However, increasing DR more than about 33.8% does not affect the planning profitability any more. This observation can be justified as follows:

In our study, the load shifting from high cost on-peak periods to low cost off-peak periods has been considered as DR that is one of the most effective demand side management approach. This kind of DR, only shifts the timing of energy use and the total energy consumption remains unchanged. However, because of the load shifting from high cost periods to low cost periods, the total cost of energy consumption is reduced. On the other hand, applying DR reduces the peak load of LDA, and would reduce the investment cost by eliminating, deferring or reducing the need for expanding new sources includes new DERs and feeders. Therefore, considering more DR percentage, improves the profitability of the planning decisions in terms of reducing both the investment and operating cost. However, this profit improvement will continue until the load profile is smoothed. After that, as there is no cost or benefit from applying the DR, increasing DR percentages does not affect the profitability of the planning decisions.

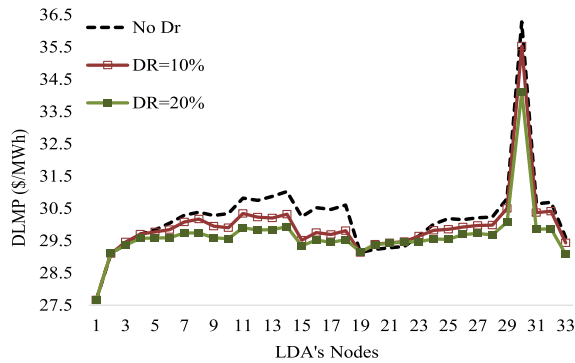


Fig. 7. Calculated DLMPs for different DR.

The calculated DLMPs as the price signals of the proposed TDSP algorithm are presented in Fig. 7 for different levels of DR. The assumed LMP changes between 19.06 \$/MWh to 42.91 \$/MWh for different load blocks. But investment on new DERs in the LDA manages the electricity price in the customer side of the network and prevents high electricity price due to the high forecasted LMP as the wholesale price of the electricity energy. Indeed, LDA's prosumers would benefit from the LDA power exchange with the transmission network in the situations of high LMPs. Increase in the DR participation level changes the DLMPs in a way to decrease the supplying cost of the LDA's customers.

IV. CONCLUSION

In this paper, a transactive DSP approach is proposed which is formulated as a maximization problem to investigate the profitability of the planning as the objective of investors in the transactive market-based environment. The proposed TDSP algorithm explores the optimal planning of the new DERs and feeders over the entire planning horizon while considering the reliable, secure and optimal operation of the LDA to satisfy the reliability and security constraints of the DSO. The modified BD is employed to divide the complicated planning problem into the IP as the master problem and the reliability and optimality problems as the corresponding sub-problems. The problem is formulated as a QCP to consider the effects of the reactive power as well as the active and reactive power losses in the TDSP utilizing the accurate AC power flow equations. The proposed algorithm calculates LDA's DLMPs to improve the input price signals for the next iteration of the TDSP process. The robust approach has been utilized to model the uncertainties associated with the forecasted errors on the renewable DERs accessible power, LMP of the interfaced transmission bus to LDA, and the LDA's load in the proposed TDSP.

The modified IEEE 33 bus distribution network in a 10-year planning horizon with 5% annual load growth has been used to provide the numerical validation for the proposed TDSP approach. The numerical results approve the capability of the proposed iterative TDSP approach to include the effects of the uncertainties on the final investors' decisions as well as the advantages of the smart DN such as demand-side management. Moreover, effects of the reactive power consumption and the

power losses (both active and reactive losses) are considered which are also reflected in the calculated DLMPs of the transactive market. This calculated transactive DLMPs are used as the input price signals for the next iteration to improve the accuracy of the robust solution. In order to benefit from the resources available at the operating stage, such as DR, only the effects of these resources are modeled in the proposed long term DSP. However, the detailed temporal constraints are not considered to make a tradeoff between accuracy and computation time. Including more detailed temporal constraints (such as ramping limits of generating units), could be considered as a suggestion for future work to study the long term expansion planning in a more realistic conditions and maintain some required chronology in dealing with short term temporal constraints.

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